

Exercise 6

Exercise 6.1 MPC stability

Consider the following MPC problem

$$\min_{u(\cdot)} V(x(T)) + \int_0^T l(x, u) dt \quad (1a)$$

$$\text{s.t. } \dot{x} = f(x, u), \quad x(0) = x_0 \quad (1b)$$

with the system dynamics function $f(x, u) = -x^3 + u$ and the integral and terminal cost $l(x, u) = qx^2 + ru^2$ and $V(x(T)) = \frac{1}{2}px(T)^2$ with positive constants p, q, r .

Using the fictitious control law $u = -kx$, what conditions must be imposed on the parameters p and k in order to satisfy the CLF inequality

$$\dot{V}(x) + l(x, -kx) \leq 0 \quad (2)$$

for all $x \in \mathbb{R}$?

If, in addition, input constraints $u \in [-1, 1]$ are taken into account, for which values of x is the CLF inequality still satisfied?

Exercise 6.2 Model predictive control with the toolbox GRAMPC

The following MPC problem with a scalar state constraint and input constraints of the form

$$\min_{u(\cdot)} V(x(T)) + \int_0^T l(x, u) dt \quad (3a)$$

$$\text{s.t. } \dot{x} = f(x, u), \quad x(0) = x_0 \quad (3b)$$

$$h(x) \leq 0 \quad (3c)$$

$$u \in [u^-, u^+] \quad (3d)$$

can be solved with the gradient-based MPC toolbox GRAMPC. In GRAMPC the constraints are considered by using outer penalty functions or the augmented Lagrangian method. In both variants, the penalty parameter can be adjusted by a heuristic method. Sketch the procedure of the numerical solution using the augmented Lagrangian method.

Add-On 1 MPC for a two dimensional crane

Consider the simplified two-dimensional crane shown in Figure 1. The trolley position is denoted by s_C , the rope length by s_R , and the load angle by ϕ .

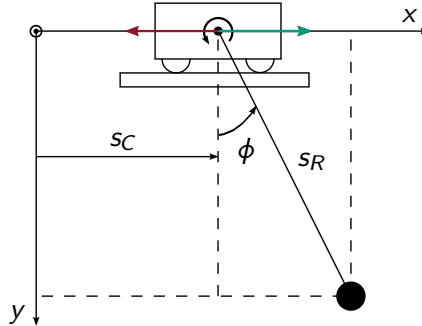


Figure 1: Schematic illustration of the 2D crane.

The angular motion of the load is described by

$$\ddot{\phi} = -\frac{1}{s_R} \left(g \sin \phi + \ddot{s}_C \cos \phi + 2\dot{s}_R \dot{\phi} \right) \quad (4)$$

where the state and input vectors are defined as

$$\mathbf{x} = [s_C \quad \dot{s}_C \quad s_R \quad \dot{s}_R \quad \phi \quad \dot{\phi}]^T, \quad \mathbf{u} = [\ddot{s}_C \quad \ddot{s}_R]^T. \quad (5)$$

The general finite-horizon MPC problem without terminal cost is

$$\min_{\mathbf{u}(\cdot)} \int_0^T l(\mathbf{x}, \mathbf{u}) dt \quad (6a)$$

$$\text{s.t. } \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{x}(0) = \mathbf{x}_0 \quad (6b)$$

$$h(\mathbf{x}) \leq 0 \quad (6c)$$

$$\mathbf{u} \in [\mathbf{u}^-, \mathbf{u}^+]. \quad (6d)$$

For moving the crane to a desired state, consider the integral cost

$$l(\mathbf{x}, \mathbf{u}) = (\mathbf{x} - \mathbf{x}_{\text{des}})^T \mathbf{Q} (\mathbf{x} - \mathbf{x}_{\text{des}}) + (\mathbf{u} - \mathbf{u}_{\text{des}})^T \mathbf{R} (\mathbf{u} - \mathbf{u}_{\text{des}}), \quad (7)$$

and limits on the angular velocity $\dot{\phi} \in [\dot{\phi}_{\min}, \dot{\phi}_{\max}]$. Additionally, the load must remain above an obstacle that is defined by the parabola

$$s_2 = -a s_1^2 - b, \quad (8)$$

where $s_1(\mathbf{x}) = x_1 + x_3 \sin(x_5)$ and $s_2(\mathbf{x}) = -x_3 \cos(x_5)$ are the coordinates of the load, and a and b are positive constants.

- Derive the state-space representation using the given state and input vectors in (5).
- Complete the provided MATLAB live script `crane_overhead.mlx` using the Symbolic Math Toolbox to define $\mathbf{f}(\mathbf{x}, \mathbf{u})$, $l(\mathbf{x}, \mathbf{u})$, and $h(\mathbf{x})$. Then use the MATLAB function `jacobian(f,x)` to analytically calculate the derivatives required for the GRAMPC problem formulation.
- Run the provided MATLAB app `Crane2D_app.m` and investigate the different constraint scenarios. Vary selected parameters, such as the prediction horizon (T_{hor}), angle limits, angular-velocity limits, or obstacle geometry, and use the comparison function to compare different parameter settings for the same scenario. Under which conditions can the crane successfully move to the desired state while avoiding the obstacle?

Add-On 2 MPC for an evasive maneuver

An application of model predictive control is the control of a single-axle vehicle as depicted in Figure 2. The position (x_1, x_2) and orientation x_3 of the vehicle define the state vector $\mathbf{x} = [x_1, x_2, x_3]^T$. The vehicle speed u_1 and the turning rate u_2 are the control inputs.

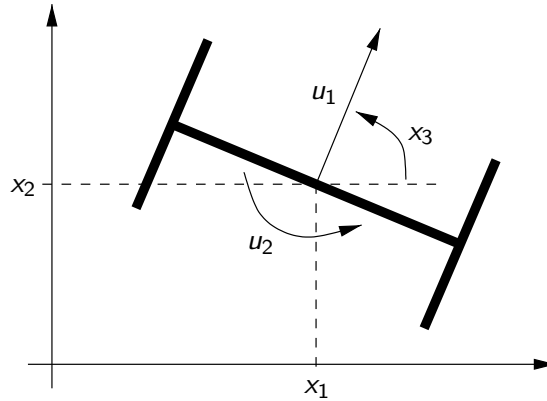


Figure 2: Schematic illustration of the single-axle vehicle.

The system dynamics of the vehicle are given by the differential equations

$$\left. \begin{aligned} \dot{x}_1 &= u_1 \cos x_3 \\ \dot{x}_2 &= u_1 \sin x_3 \\ \dot{x}_3 &= u_2 \end{aligned} \right\} \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad (9)$$

where the control inputs are constrained by

$$u_1 \in [-20 \text{ m s}^{-1}, 20 \text{ m s}^{-1}], \quad u_2 \in [-1 \text{ rad s}^{-1}, 1 \text{ rad s}^{-1}]. \quad (10)$$

The control scenario is to perform an evasive maneuver from the initial position

$$\mathbf{x}_0 = [0 \text{ m} \quad 0 \text{ m} \quad 0 \text{ rad}] \quad (11)$$

to the target position

$$\mathbf{x}_f = [40 \text{ m} \quad 0 \text{ m} \quad 0 \text{ rad}], \quad (12)$$

while preventing a collision with a pedestrian at the position $(x_{p,1}, x_{p,2}) = (20 \text{ m}, 1 \text{ m})$. To this end, the vehicle must maintain a minimum distance of 3 m to the pedestrian.

- Formulate a suitable optimization problem including the constraints to evade the pedestrian.
- In order to prepare for the programming tutorial, where GRAMPC is used to solve the problem, calculate all analytical gradients that must be provided in the problem formulation for GRAMPC.
- Under which conditions could the evasion fail?